Amira Hijazi

NSF AI Institute for Advances in Optimization (AI4OPT)
Georgia Institute of Technology
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EDUCATION

Ph.D. in Industrial and Systems Engineering 2018 – 2023
North Carolina State University, Raleigh, NC
- Thesis: Resource Allocation Using Mathematical Programming and Machine Learning
- Advisors: Osman Ozaltin and Reha Uzsoy

B.S. in Industrial Engineering 2012 – 2016
Islamic University of Gaza, Gaza, Palestine
- Final Project: Proposed Scheduling Models Based on Real-Life Instances

PREPRINTS AND WORKING PAPERS

Underlined indicates students under my supervision

[1] End-to-End Supply Chain Planning in the Paper Industry Via Column Generation and Benders Decomposition.
Changkun Guan, **Amira Hijazi**, Pascal Van Hentenryck. *Under Review.*

[2] Practice-Based Optimization for the Strategic Locomotive Assignment Problem.
Yunji Kim, **Amira Hijazi**, Kevin Dalmeijer, Pascal Van Hentenryck.

[3] Route Optimization Over Scheduled Services For Large-Scale Package Delivery Networks.
Mohammed Faisal Ahmed, **Amira Hijazi**, Pascal Van Hentenryck, Ahmed El Nashar.

[4] Decision-Focused Learning for Pickup and Delivery Problems with Time Windows, Split Load, and Transshipment.
Sikai Cheng, **Amira Hijazi**, Pascal Van Hentenryck, Alan Erera.

JOURNAL PUBLICATIONS

Underlined indicates students under my supervision during the paper work.

[1] An Improving Column is All You Need: Enhancing Column Generation for Parallel Machine Scheduling via Transformers.
Amira Hijazi, Osman Ozaltin, Reha Uzsoy. *INFORMS Journal on Computing*, June 2026.

[2] Contextual Stochastic Optimization for Omnichannel Multi-Courier Order Fulfillment Under Delivery Time Uncertainty.
Tinghan Ye, Sikai Cheng, **Amira Hijazi**, Pascal Van Hentenryck. *Manufacturing & Service Operations Management*, October 2025.
- 2025 INFORMS MSOM Practice-Based Research Competition, Honorable Mention.

CONFERENCE PROCEEDINGS AND EXTENDED ABSTRACTS

- Locomotive Assignment under Schedule Disruptions.
  Yunji Kim, **Amira Hijazi**, Kevin Dalmeijer, Pascal Van Hentenryck.
  INFORMS Transportation Science and Logistics Society Conference, July 2026.

- SPOT: Spatio-Temporal Pattern Mining and Optimization for Load Consolidation in Freight Transportation Networks.
  Sikai Cheng, **Amira Hijazi**, Jeren Konak, Alan Erera, Pascal Van Hentenryck. *IEEE International Conference on Data Mining*, November 2025.
  - Conference Acceptance Rate is 13%

- Conformal Predictive Distributions for Order Fulfillment Time Forecasting.
  Tinghan Ye, **Amira Hijazi**, Pascal Van Hentenryck. *International Conference on Computational Logistics*, September 2025.

- Confidence-Aware Deep Learning for Load Plan Adjustments in the Parcel Service Industry.
  Thomas Bruys, Reza Zandehshahvar, **Amira Hijazi**, Pascal Van Hentenryck. *AAAI-25 Bridge: AI+ORMS Workshop.*

RESEARCH EXPERIENCE

Georgia Institute of Technology Atlanta, GA
NSF AI Institute for Advances in Optimization (AI4OPT)
Research Engineer II, AI Lead for Supply Chains Thrust Sep 2024–Present
Leading multiple research projects in collaboration with UPS, Union Pacific, Smurfit Westrock, and Best Buy to address large-scale supply chain problems using AI and optimization.
Projects: Route Optimization For Large-Scale Package Delivery Networks.
Data-Driven E-Commerce Robust Order Fulfillment with Delivery Uncertainty
Integrated Large-Scale Lot Sizing, Cutting Stock, and Transportation Planning
Strategic Locomotive Assignment Optimization
Constraint-aware Pattern Mining for Dynamic Load Consolidation
End-to-End Learning for Parallel Machine Scheduling

Postdoctoral Fellow Jul 2023 – Sep 2024

North Carolina State University Raleigh, NC
Research Assistant, Department of Industrial and Systems Engineering Aug 2020 – Jun 2023
Projects: Learning-based optimization methods for production scheduling problems
Risk-Averse Vaccine Allocation Using Stochastic Programming

Los Alamos National Laboratory Los Alamos, NM
Research Intern, Theoretical Division Jun 2020 – Sep 2020
Project: Correcting Misalignment in Drone-Mounted LiDAR Point Clouds

European Gaza Hospital Gaza, Palestine
Data Analytics Intern Jun 2015 – Aug 2015

TEACHING EXPERIENCE

North Carolina State University Raleigh, NC
Teaching Assistant, Department of Industrial and Systems Engineering Aug 2019 – May 2020

Course: Quality Design and Control, Undergraduate Level

Islamic University of Gaza Gaza, Palestine
Teaching Assistant, Department of Industrial Engineering Aug 2016 – Oct 2016
Courses: Operations Research, Operations Management, Statistical Quality Control, Materials Engineering, Systems Simulation, Engineering Economy

PHD STUDENTS SUPERVISION

1. Changkun Guan

Advisor:	Pascal Van Hentenryck
Degree Program:	Operations Research
Project:	*Large Scale Supply Chain Optimization for Paper Manufacturing Industry*
Advising Dates:	2025–Present

2. Tinghan Ye

Advisor:	Pascal Van Hentenryck
Degree Program:	Industrial Engineering
Project:	*Data-Driven E-Commerce Robust Order Fulfillment with Delivery Uncertainty*
Advising Dates:	2023–Present
Activities:	Co-author on 1 journal and 1 conference article
	Presenter at 3 conferences
Award:	INFORMS MSOM Practice-Based Research Competition, Honorable Mention

3. Yunji Kim

Advisor:	Pascal Van Hentenryck
Degree Program:	Industrial Engineering
Project:	*Strategic Locomotive Assignment Optimization*
Advising Dates:	2023–Present
Activities:	Co-author on 1 submitted article
	Presenter at 2 conferences

4. Sikai Cheng

Advisor:	Pascal Van Hentenryck
Degree Program:	Machine Learning
Projects:	*Data-Driven E-Commerce Robust Order Fulfillment with Delivery Uncertainty*
	Dynamic Load Consolidation For Large-Scale Package Delivery Networks
Advising Dates:	2023–Present
Activities:	Co-author on 1 journal and 1 conference article
	Presenter at 2 conferences
Award:	INFORMS MSOM Practice-Based Research Competition, Honorable Mention
	Paper Accepted at IEEE Data Mining Conference with Acceptance Rate 13%

5. Jeren Konak

Advisor:	Alan Erera
Degree Program:	Operations Research
Project:	*Dynamic Load Consolidation For Large-Scale Package Delivery Networks*
Advising Dates:	2024–2025
Activities:	Co-author on 1 conference article
	Presenter at 1 conference
Award:	Paper Accepted at IEEE Data Mining Conference with Acceptance Rate 13%

6. Mohammed Faisal Ahmed

Advisor:	Pascal Van Hentenryck
Degree Program:	Machine Learning
Project:	*Route Optimization For Large-Scale Package Delivery Networks*
Advising Dates:	2024–Present
Activities:	Presenter at 1 conference

7. Thomas Bruys

Advisor:	Pascal Van Hentenryck
Degree Program:	ECE Department
Project:	*Deep Learning for Load Plan Adjustments in the Parcel Service Industry*
Advising Dates:	2023–2025
Activities:	Co-author on 1 submitted article Presenter at 2 conferences

AWARDS & HONORS

INFORMS MSOM Practice-Based Research Competition, Honorable Mention	2025
College of Engineering Travel Award for INFORMS Annual Meeting ($500)	Oct 2019
College of Engineering Travel Award to attend "Understanding Health Behavior using Smartphones and Wearables Bootcamp" in Saskatoon, Canada	Jun 2019
Recipient of University Graduate Fellowship ($4000)	Aug 2018
College of Engineering Travel Award for INFORMS Annual Meeting ($500)	Nov 2018
Hani Qaddumi B.S. Scholarship, Hani Qaddumi Foundation	Aug 2015 - May 2016
Islamic University of Gaza Academic Excellence Scholarship	Aug 2012 - May 2016

CONFERENCE AND WORKSHOP PRESENTATIONS

Underlined: Student co-author, ∓: Presenting co-author

- Amira Hijazi, "AI and Optimization for Supply Chain and Logistics," 2026 Faculty Training Program Conference INFORMS Annual, AI4OPT, Georgia Tech, Atlanta, GA, June 2026
- Thomas Bruys∓, Reza Zandehshahvar, Amira Hijazi, Pascal Van Hentenryck, "Uncertainty-Aware Load Planning with Deep Learning for the Parcel Service Industry," 2025 INFORMS Annual Meeting, Atlanta, GA, October 2025
- Tinghan Ye∓, Sikai Cheng, Amira Hijazi, Pascal Van Hentenryck, "Contextual Stochastic Optimization for Omnichannel Multi-Courier Order Fulfillment Under Delivery Time Uncertainty," 2025 INFORMS Annual Meeting, Atlanta, GA, October 2025
- Sikai Cheng∓, Amira Hijazi, Jeren Konak, Alan Erera, Pascal Van Hentenryck, " Spatio-Temporal Pattern Mining and Optimization for Load Consolidation in Freight Transportation Networks," 2025 INFORMS Annual Meeting, Atlanta, GA, October 2025
- Yunji Kim∓, Amira Hijazi, Kevin Dalmeijer, Pascal Van Hentenryck, "Practice-Based Optimization for Strategic Locomotive Assignment Problem," 2025 INFORMS Annual Meeting, Atlanta, GA, October 2025
- Jeren Konak∓, Amira Hijazi, Sikai Cheng, Alan Erera, "Improving Linehaul Efficiency with Load Consolidation Models", 2025 INFORMS Annual Meeting, Atlanta, GA, October 2025

- Tinghan Ye∓, Amira Hijazi, Pascal Van Hentenryck, "Conformal Predictive Distributions for Order Fulfillment Time Forecasting," 2025 International Conference on Computational Logistics, Rotterdam, Netherlands, September 2025
- Tinghan Ye, Sikai Cheng, Amira Hijazi, Pascal Van Hentenryck∓, "Contextual Stochastic Optimization for Omnichannel Multi-Courier Order Fulfillment Under Delivery Time Uncertainty," 2025 INFORMS Manufacturing and Service Operations Management Conference, London, United Kingdom, June 2025
- Amira Hijazi∓, Osman Ozaltin, Reha Uzsoy, "Enhancing Column Generation for Parallel Machine Scheduling via Transformers", 2025 IISE Annual Conference, Atlanta, GA, June 2025
- Thomas Bruys∓, Reza Zandehshahvar, Amira Hijazi, Pascal Van Hentenryck, "Uncertainty-Aware Load Planning with Deep Learning for the Parcel Service Industry," AAAI 2025 Bridge Program on Combining AI and OR/MS for Better Trustworthy Decision Making, Philadelphia, Pennsylvania, February 2025
- Amira Hijazi, "AI and Optimization for Supply Chains,", GaTech AI4OPT Faculty Training Program, Atlanta, GA, June 2024
- Amira Hijazi∓, Osman Ozaltin, Reha Uzsoy, "Neural Network and Column Generation for Parallel Machine Scheduling," 2022 INFORMS Annual Meeting, Indianapolis, IN, October, 2022
- Amira Hijazi∓, Osman Ozaltin, Maria Jensen, Hamed Yarmand, Julie Ivy, "Risk-Averse Vaccine Allocation Using Stochastic Programming," Virtual INFORMS Annual Meeting, 2020
- Amira Hijazi∓, Osman Ozaltin, Maria Jensen, Hamed Yarmand, Julie Ivy,"Mean-Risk Two-Stage Vaccine Allocation to Different Regions Under Uncertainty", 2019 INFORMS Annual Meeting, Seattle, WA, October 2019

SERVICE & LEADERSHIP

Journal ad-hoc Reviewer
Manufacturing & Service Operations Management
Computers & Industrial Engineering
Journal of Heuristics

HerWILL AI for Digital Safety Global Datathon — Apr 2026
Judge

INFORMS ORMS Tomorrow Student Magazine
Mini Poster Competition Judge — Dec 2025
Co-Lead Editor — Apr 2020 - Sep 2021

IISE Annual Conference — May 2025
Reviewer and Session Chair

NC State University
ISE Department: DEI Committee Graduate Students Representative — Oct 2020 - Aug 2021
EMPOWER Collaborative: Student Ambassador — Aug - Oct 2020
Pursuing Operational Excellence Task Force: Student Representative — Dec 2019 - Jul 2020
Graduate Student Association: ISE Department Representative — Sep 2019 - May 2021
Graduate Student Association: Research Recognition Committee Member — Sep 2019 - May 2020

INFORMS Chapters/Fora Committee — Feb 2020 - Feb 2021
Student Chapter Representative

INFORMS Health Applications Society — Feb 2020 - Dec 2020
Student Liaison

INFORMS Annual Meeting
Daily E-news Student Writer — Oct 2019
Student Volunteer — Nov 2018

Islamic University of Gaza IISE Student Chapter — Aug 2013 - May 2015
Founder and President

Arab Innovation Network — Jun 2013 - Jun 2015
Student Ambassador